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Special Issue

A Versatile Urea Type Linker for Functionalizing Natural Glycans and Its Validation in Glycan Arrays

Sonia Serna,*[a] Raik Artschwager,*[a, d] Damián Pérez-Martínez,[a] Rosa Lopez,[b] and Niels-Christian Reichardt[a, c]

The isolation from organisms and readily available glycoproteins has become an increasingly convenient source of N-glycans for multiple applications including glycan microarrays, as reference standards in glycan analysis or as reagents that improve bioavailability of protein and peptide therapeutics through conjugation. A problematic step in the isolation process on a preparative scale can be the attachment of a linker for the improved purification, separation, immobilization and quantification of the glycan structures. Addressing this issue, we firstly aimed for the development of an UV active linker for a fast and reliable attachment to anomeric glycosylamines via urea bond formation. Secondly, we validated the new linker on glycan arrays in a comparative study with a collection of N-glycans which were screened against various lectins. In total, we coupled four structurally varied N-glycans to four different linkers, immobilized all constructs on a microarray and compared their binding affinities to four plant and fungal lectins of widely described specificity. Our study shows that the urea type linker showed an overall superior performance for lectin binding and once more, highlights the often neglected influence of the choice of linker on lectin recognition.

Introduction

Protein glycosylation is a highly complex and essential post-translational modification that decorates the cell membranes with a dense multivalent layer of glycans. Cell surface glycans are covalently connected either to lipids (e.g. via phosphate ester bonds) or proteins via serine or threonine residues (O-glycans), asparagine (N-glycans) or hydroxylysine or -proline (e.g. collagens). Many signalling processes are initiated by the interaction of carbohydrates and their recognizing protein counterparts, known as glycan binding proteins (GBP) or lectins.[1] These interactions are crucial in processes including pathogen recognition with subsequent immune activation, cell adhesion and migration and in apoptosis among others. Consequently, techniques that allow fast and reliable characterization of carbohydrate-lectin interactions are of a great importance in glycomics. In this context, glycan microarray technology has revolutionized the field and has become an essential technique for the high throughput study of binding affinities GBPs and glycans.[2–5] However, hampered by the access to large libraries of pure glycan structures it remains difficult to take full advantage of the available glycan array platforms. Robust access to well characterized complex glycans in sufficient amounts is granted by chemo-enzymatic synthesis but this approach is time consuming and laborious and requires highly experienced personnel. On the other hand, natural occurring glycans are an alternative source to rapidly populate glycan arrays with a large variety of glycan structures.[6] The low molecular weight, high hydrophilicity and low reactivity of the carbohydrates however hampers an efficient immobilization on activated microarray slides.[7] Although direct printing of free reducing sugars on hydrazine activated slides has been described,[8] it is generally accompanied by low immobilization rates, requires a high printing glycan concentration and results in hydrazone bonds with limited stability. In addition, the direct immobilisation of free reducing glycans is usually limited to the use of synthetic substrates, as natural glycans are commonly fractionated by chromatography after tagging with a UV sensitive tag. Hence, the immobilization of natural glycans on surfaces benefits largely from a previous functionalization with hetero bifunctional and UV-active linkers. A suitable linker should not change the closed ring conformation at the reducing end but serve as a "catch and release"[9] and/or immobilisation anchor.

Most current methods for labelling of natural glycans with a UV-active tag or linker are based on reductive amination of the reducing end of the carbohydrates.[10–12] Although efficient functionalization and immobilization is achieved by these protocols, the loss of the natural conformation in the anomeric position can impede or even abolish the efficient recognition

[a] S. Serna, R. Artschwager, D. Pérez-Martínez, N.-C. Reichardt
Glycotechnology Group
Basque Research and Technology Alliance (BRTA)
CIC biomaGUNE
Paseo Miramon 194, 20014 Donostia-San Sebastián (Spain)
E-mail: nreichardt@cicbiomagune.es

[b] R. Lopez
Organic Chemistry Department I
University of the Basque Country (UPV/EHU)
Paseo Manuel Lardizabal 3, 20018, Donostia-San Sebastián (Spain)

[c] N.-C. Reichardt
CIBER-BBN
Paseo Miramon 194, 20014 Donostia-San Sebastián (Spain)

[d] R. Artschwager
Current address: Memorial Sloan Kettering Cancer Center
New York, New York 10065 (USA)

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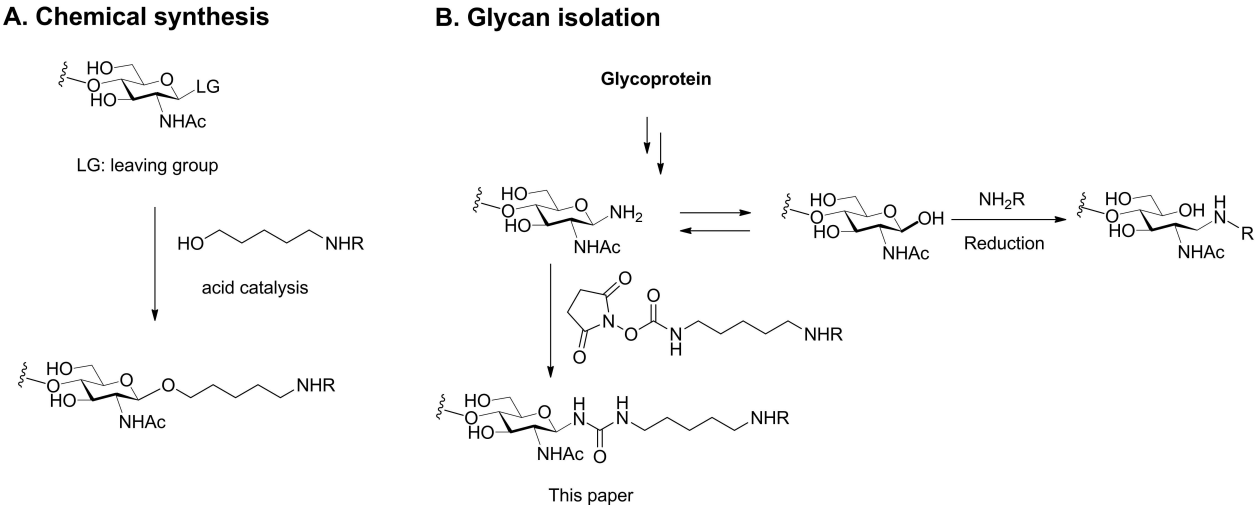
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by lectins especially for smaller glycan structures. Various other methods have been described for example using glycosylamines[13,14] and hydroxylamines[11,15–17] that maintain the cyclic ring structure with different linker aiming for an improved purification and screening of glycans of interest, but these reactions are reversible in acidic environments limiting their broader application.

Glycan presentation is an additional important aspect for the choice of a linker and here glycopeptides are an obvious choice as attachment via serine a threonine or asparagine (Asn) residue mimics the natural presentation in a glycan binding protein. While solid phase peptide synthesis allows the preparation of large peptide libraries, their preparation is quite limited by the high cost and reduced reactivity of the Fmoc protected glycosyl amino acids, especially in the case of larger N-glycan structures.[18] More readily accessible asparagine functionalized N-glycans can be prepared via Pronase® digestion of glycoproteins and, for example, temporally Fmoc protected for purification and deprotected to release an amine group for NHS activated surface immobilization.[19–21] In certain cases, such as the study of MUC1 glycopeptides as cancer markers, a natural peptide linker is crucial as the recognition by antibodies involves both glycan and peptide parts.[22]

Historically, most glycans made for microarray-based studies had been prepared by chemical or chemo-enzymatic synthesis which facilitates the early introduction of a linker in the synthetic strategy. In general terms, glycoconjugates and carbohydrates prepared by chemical synthesis are produced by glycosylation of primary alcohols with activated glycosidic donors. (Scheme 1) For glycan arrays, the choice of surface chemistry and linker will influence the degree of immobilization and consequently the glycan surface concentration which in turn affects the orientation and accessibility of the printed glycan.[23] In general, carbohydrate structures immobilized via linear synthetic linkers are likely to provide higher flexibility and consequently engage in effective lectin interactions with higher probability. Also, the dense carbohydrate immobilization results in a multivalent presentation leading generally to increased avidity that is influenced by the distance between individual glycan molecules on the surface.[24] Combining glycans obtained from natural sources with synthetic glycans often resulted in a mix of different linkers employed on the same array implicating differences in the presentation of individual glycans on the arrays.[23] In previous studies aimed at comparing different microarray formats only small variations in lectin binding were seen likely due to different glycan surface densities which are more relevant for weak carbohydrate-lectin interactions.[25,26] Others have reported that a lack of interaction in microarrays could be correlated with the structure of the employed linker,[27] as certain linkers without being part of the binding epitope can lead the glycan to an unfavourable orientation.[28,29] Our group has made an important effort to prepare a library of N-glycan structures by chemical and chemo-enzymatic synthesis equipped with a pentylamino linker, that had been widely used in many synthetic glycan arrays at that time.[30] To complement this N-glycan library with readily available natural N-glycans, we envisaged their functionalisation with an urea based linker inspired by fluorescent tags routinely employed in the chromatographic analysis of peptides and glycans.[31,32] Therefore, we firstly, aimed for the functionalization of glycosylamines with an NHS activated carbamate linker producing a stable urea linkage that preserves the closed ring conformation of the carbohydrates and allows for purification purposes (Scheme 1).[31] Secondly, to evaluate the influence of linkers on the recognition of different N-glycan epitopes in more detail we systematically studied various glycan lectin interactions for 4 structures immobilized via 4 different linkers as length and nature of the spacer have been described to be a key parameter influencing the binding of lectins to glycan microarrays.[23,28,29]



Scheme 1. General approaches towards linker functionalized carbohydrates. A. Multistep chemical synthesis, O-glycosylation with primary alcohols. B. Isolation from natural sources, open ring functionalization based on imine formation/reduction and the proposed strategy to functionalize glycosylamines with urea linker presented herein.

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Results and discussion

Linker synthesis

The synthetic design of the linker should fulfil the following key characteristics: (i) the presence of a chromophoric/hydrophobic functionality to facilitate detection and separation upon purification of natural glycans, (ii) efficient chemical synthesis on a multigram scale and (iii) contain a masked amino pentyl moiety for immobilization on arrays or proteins and nanoparticles. We synthesized three linkers with *N*-carboxybenzyl (Cbz), *N*-benzyl-*N*-carboxybenzyl (benzyl CBz) and 2-naphthylmethyl carbamate (CNAP) as chromophores, and selected CNAP for its good UV absorption and rapid and clean cleavage by hydrogenolysis (a comparative study of three UV active linkers at equimolar concentrations (100 μM) is shown in Figure S1). The CNAP protected linker **6** was readily available on a multigram scale in 5 steps. First, 2-hydroxymethylnaphtalene **1** was converted to the CNAP protected aminopentanol **2** (Scheme 2). The alcohol was then transformed to the corresponding mesylate **3**, which was displaced by an azide to arrive at **4** in high yield. After Staudinger reduction of the azide, the free amine **5** was converted to the respective NHS activated compound **6** by treatment with disuccinimidyl carbonate (DSC) in excellent yield over the two steps.

Glycan isolation and derivatization

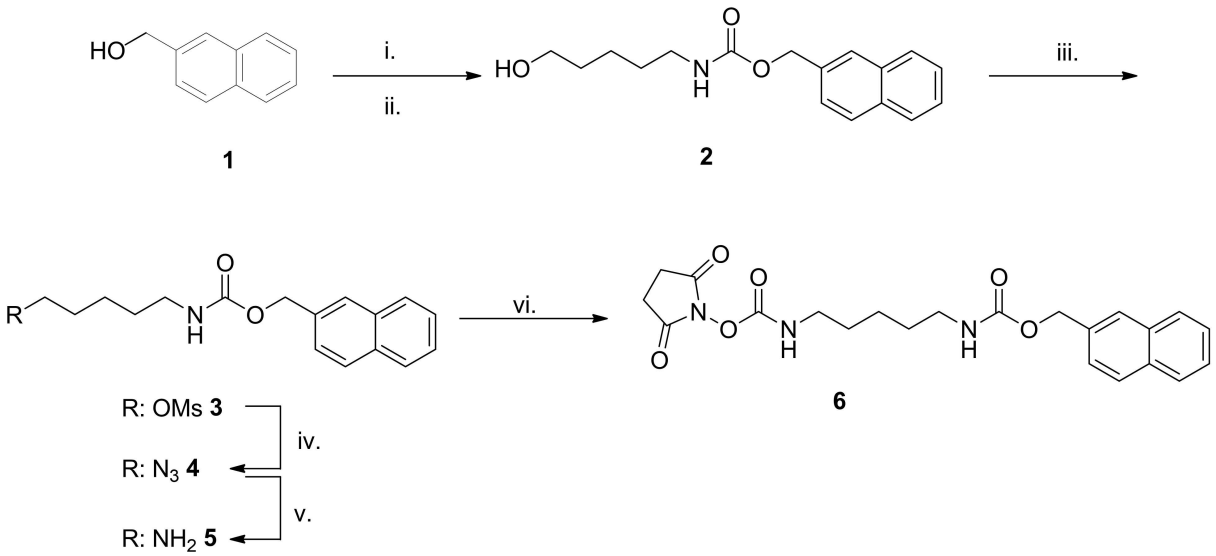
Glycan isolation

Sialylglycopepide (SGP) was isolated from egg yolk powder and purified to homogeneity. (Scheme 3).[33] N-acetyl-neuraminic acids were removed by mild acidic treatment as described previously.[34] This treatment produced bis galactosylated biantennary glycopeptide (G2-peptide) along with a variable content (ranging from 16% to 33%) of mono galactosylated biantennary glycopeptide (G1-peptide). This G2/G1-peptide mixture **7** served as a common intermediate for further synthesis of different N-glycan structures. The glycan was cleaved from the peptide backbone by treatment with PNGaseF and the corresponding hemiacetal **8** was isolated in 94% of yield.

Linker functionalization

Although N-glycan removal with PNGaseF produces an intermediary glycosylamine ready for direct conjugation with an activated linker, we decided to hydrolyse this unstable intermediate to demonstrate a general applicability of our method to free reducing glycans. The (re)introduction of the glycosylamine was carried out by Kochetkov amination through treatment with $(NH_4)HCO_3$[35] and was further reacted with NHS activated linker **6** and DIPEA in DMSO at room temperature. As the activated linker **6** was prone to suffer hydrolysis during the conjugation reaction, sequential addition of linker was found to be advantageous. Increasing the reaction temperature to 50 °C and 80 °C, respectively, resulted in lower yields likely due to the decomposition of the NHS activated ester at elevated temperatures. After the isolation of linker functionalized N-glycan **10** (Figure S2), NMR analysis confirmed the formation of an anomeric urea bond with β-conformation evident from the coupling constant of $J = 9.5$ Hz between H-1 and H-2. Finally, the CNAP carbamate group was removed by catalytic hydrogenolysis in quantitative yield.[36]

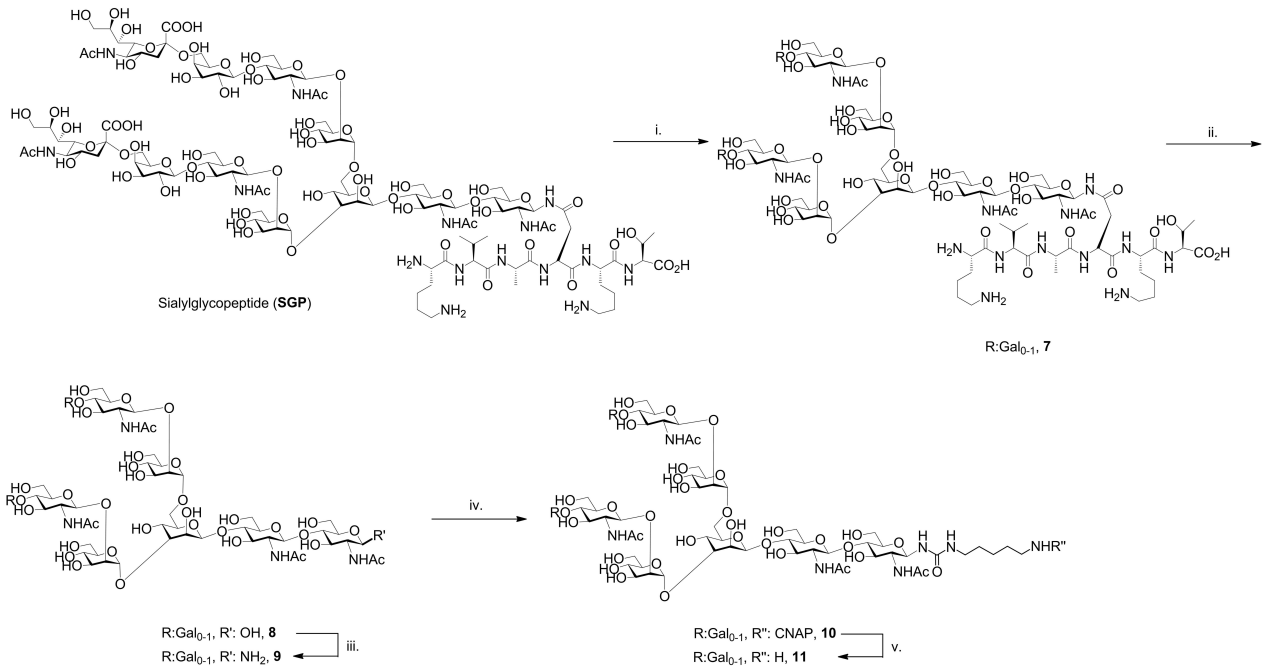
From the galactosylated N-glycan (**G1/G2**) linker conjugate **11** we produced a small number of N-glycans by treatment with several glycosyltransferases and glycosidases as summarized in Scheme 4. Treatment of **11** with recombinant bovine



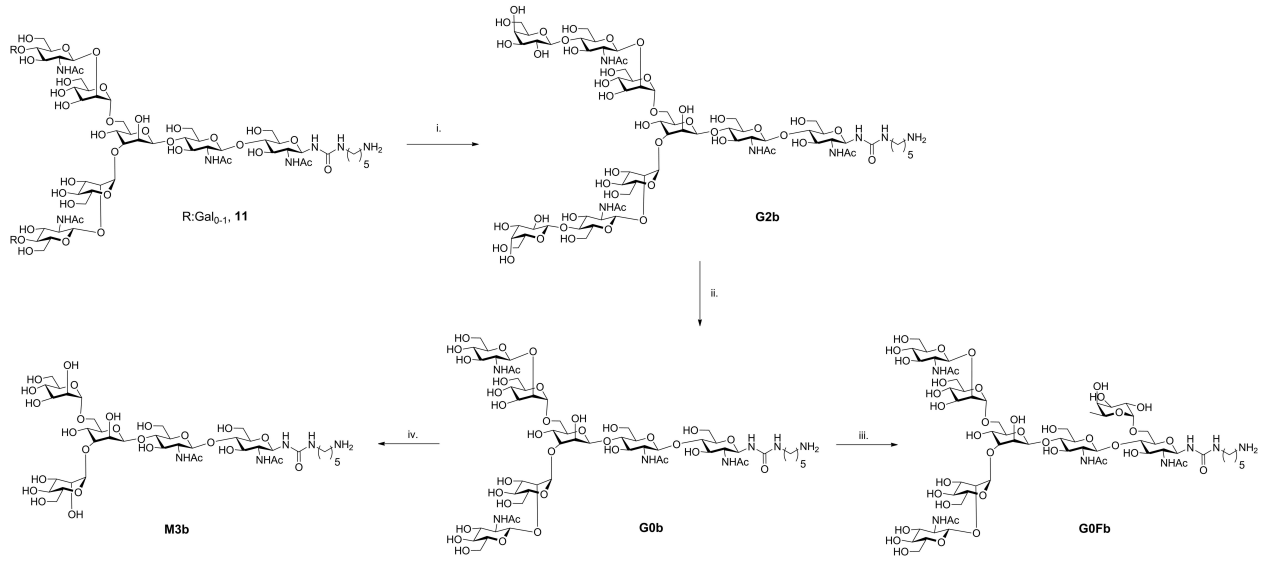
Scheme 2. Synthesis of NHS activated linker. Reagents and conditions: i. *p*-nitrophenylchloroformate, pyridine, CH_2Cl_2. ii. 5-amino-1-pentanol, NEt_3, CH_2Cl_2, (85%, over two steps) iii. MsCl, NEt_3, CH_2Cl_2, (90%) iv. NaN_3, DMF, 40 °C (98%); v. PPh_3, THF:H_2O (10:1) (95%); vi) DSC, DIPEA, CH_2Cl_2, 50 °C (96%).

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Scheme 3. Isolation of G1/G2 N-glycan from egg yolk powder and functionalization with linker **i**. AcOH, 80 °C, quantitative,[34] **ii**. PNGase F, 20 mM Tris, 50 mM NaCl, 5 mM EDTA, pH 7.5, 37 °C (94%); **iii**. NH_4HCO_3, water, 30 °C; **iv**. Linker **6**, DIPEA, DMSO, rt (69%); **v**. H_2, 10% Pd/C, EtOH/water (1:1) containing 1% TFA, 50 °C (99%).



Scheme 4. Enzymatic modifications of isolated N-glycan structures. Reagents and conditions: **i**. Bovine milk β-1,4-galactosyltransferase, UDP-galactose, 20 mM $MnCl_2$, Tris 50 mM pH 7.5, 37 °C (83%) **ii**. β-Galactosidase from *Aspergillus niger*, NaOAc buffer (100 mM) pH 4.5, 37 °C (91%) **iii**. Core α-1,6-fucosyltransferase from *Homo sapiens* (FUT8), GDP-fucose, Tris 100 mM pH 7.4, rt (86%). **iv**. β-N-acetylglucosaminidase S, 20 mM $CaCl_2$, NaOAc buffer (50 mM) pH 5.5, 37 °C (95%).

milk β-1,4-galactosyltransferase and UDP-GlcNAc produced homogeneous **G2b**, while the **G0b** derivative, with terminal GlcNAc residues was obtained by enzymatic hydrolysis of the galactose residues with β-galactosidase from *Aspergillus niger*. The core α-1,6-fucosylated N-glycan **G0F** derivative was obtained by incubation of **G0b** with recombinant human α-1,6-fucosyltransferase FUT8 in the presence of GDP-fucose. Paucimannose derivative (**M3b**) was obtained from the respective **G0b** precursor by treatment with β-N-acetylglucosaminidase S from *Streptococcus pneumoniae* to produce the common core pentasaccharide structure Man3.

In addition, the sequence of enzymatic modifications described in Scheme 4 was applied to modify N-glycans functionalized with the peptide (KVANKT) at the reducing end (See Supporting Information, Scheme S1) to arrive at compounds **G0c, G2c, M3c** and **G0Fc** respectively. Moreover, the

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treatment of the N-glycopeptides **G0c, G2c, M3c** and **G0Fc** with Pronase produced the glycosyl asparagines **G0d, G2d, M3d** and **G0Fd** by cleaving all peptide bonds.

In summary, the chemo-enzymatic derivatization of isolated SGP allows the high yielding synthesis of **G0**, **G2**, **M3** and **G0F** N-glycans linked to the natural egg derived peptide (KVANKT), Asn linker[37] and the urea type linker developed for the functionalization of natural glycans. Additionally, 5-aminopentyl linker modified N-glycan structures (**G2a**, **G0a**, **M3a** and **G0Fa**) which had been previously synthesized [30][38] by a chemo-enzymatic strategy were included as part of our study to evaluate the effect of the different linkers on glycan presentation on the microarrays (Table 1).

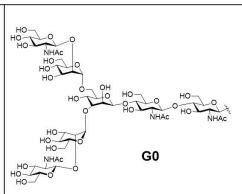
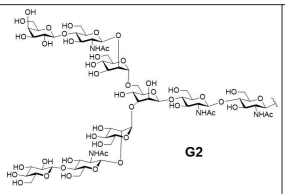
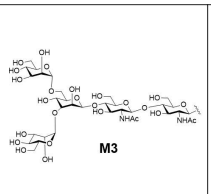
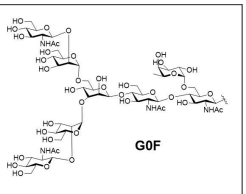
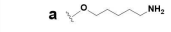
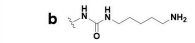
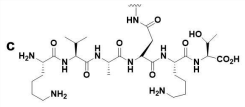
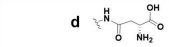
Glycan microarray preparation

Glycan microarrays were prepared by the covalent immobilisation of glycoconjugates carrying amino groups onto NHS activated glass slides with the help of a printing robot. N-glycans functionalized with the synthetic linkers **a** and **b** have a terminal primary amino group which reacts efficiently at basic printing pH. The asparagine modified N-glycans also contain a single free amino group and despite of the short length of this linker, it has been described to orient the carbohydrates perpendicular to the microarray surface which is considered a favourable conformation for GBP recognition.[28] On the other hand, the glycopeptide sequence (KVANKT) contains three possible anchoring points. We assumed that the printing pH of 8.5 will allow the N-terminal α-amino group with a pKa of 7.7 to react with the NHS-activated surface rather than the lysine residues in the peptide sequence with a higher pKa of 10.5.[39] All glycans were printed at identical concentration to facilitate a homogeneous distribution and immobilization of the carbohydrates on the spots. The glycan printing concentration was below the surface concentration of reactive NHS groups, which should at least in theory allow complete immobilization of the spotted glycans leading to equal amounts of carbohydrates immobilized per spot.[40]

Lectin incubation

In order to evaluate the linker influence in lectin-glycan recognition, glycan microarrays were interrogated with plant and fungal lectins with known binding specificities.[41] Commercially available lectins were conjugated with Alexa Fluor™ 555 NHS activated ester, and the degree of labelling was adjusted to avoid the rapid saturation of the binding assays.[42] Many glycan microarray studies rely on the incubation of GBPs at a single concentration and the interaction strength with different printed glycans is ranked considering the averaged RFU values. It has been shown ranking glycans binding at a single concentration does not necessarily correlate with the ranking of affinity constants calculated from data obtained at several different GBP concentrations.[43] For this reason, glycan microarrays were interrogated with labelled lectins at 7 different concentrations, which had been established experimentally for each lectin to ensure a large dynamic range with saturation occurring only at the most concentrated solutions. The chosen lectins recognize the following structural elements in N-glycans as follows: *Concanavalin A* is a mannose/glucose binding lectin, interacting with the central trimannosidic core presented in all N-glycan structures; *Griffonia (Bandeiraea) Simplicifolia* lectin II (BSL-II) selectively binds to terminal GlcNAc residues in **G0** and **G0F** structures; *Erythrina cristagalli* agglutinin (ECA) recognizes efficiently Galβ1-4GlcNAc in **G2** structures; *Aleuria aurantia* lectin (AAL) binds to fucosylated oligosaccharides and wheat germ agglutinin (WGA) binds terminal GlcNAc residues and the diacetyl-chitobiose motif common to all N-glycans. Figure 1 shows a summary of the lectin binding profiles for all glycoconjugates in histogram format.

Table 1. Collection of 16 N-glycan structures included in the glycan microarrays.

Linker \ *N-glycan*	 G0	 G2	 M3	 G0F
a 	**G0a**	**G2a**	**M3a**	**G0Fa**
b 	**G0b**	**G2b**	**M3b**	**G0Fb**
c 	**G0c**	**G2c**	**M3c**	**G0Fc**
d 	**G0d**	**G2d**	**M3d**	**G0Fd**

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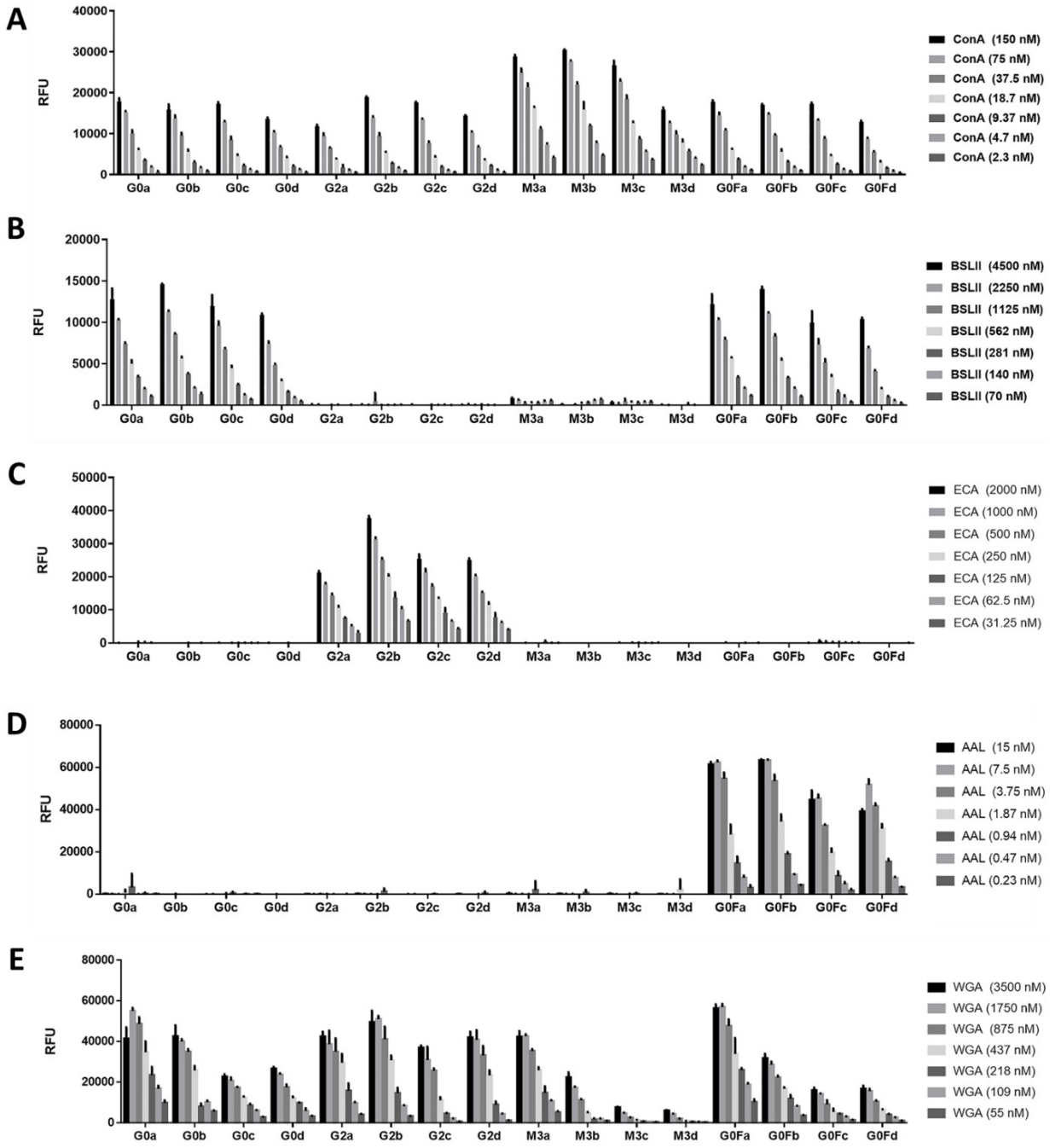


Figure 1. Binding profile of GBPs on N-glycan microarrays. Each histogram represents the average RFU of six spots along with the standard deviation of the mean. Microarrays contained four general N-glycans (**G0**, **G2**, **M3** and **G0F**) immobilized with four different linkers (**a**–**d**) A. *Concanavalin A* (ConA, α-Man/Glu) B. *Griffonia simplicifolia* lectin II (BSLII, terminal GlcNAc) C. *Erythrina crystagalli* agglutinin (ECA, Galβ1-4GlcNAc) D. *Aleuria aurantia* lectin (AAL, Fuc) E. Wheat germ agglutinin (WGA, chitobiose).

For a more systematic study of the influence of the linker on lectin binding, the data sets were subjected to standard Pearson correlation analysis. Pearson correlation analyses the linearity of binding intensities between two chosen linkers for all glycans interacting with a particular lectin (Figure 2).

Good correlation was observed in general for different pairs of linkers (Figure 2). In particular, our novel urea type linker **b** showed an excellent correlation with the "natural" peptidic linkers **c** and **d** (*r*: from 0.93 to 1, Figure 2D–E). Moreover, N-glycan asparagines (linker **d**) bound with nearly equal intensity as N-glycans immobilized via the larger peptidic linker **c** (Figure 2F). *Concanavalin A* (ConA), *Erythrina cristagalli* agglutinin (ECA) and *Aleuria aurantia* lectin (AAL) showed excellent correlation across the linker pairs with *r* values ranging from 0.90 to 1. On the other hand, *Griffonia* (*Bandeiraea*) *Simplicifolia* Lectin II (BSL-II) and wheat germ agglutinin (WGA) showed much lower correlation for certain linker pairs. BSL-II lectin is known to recognize terminal GlcNAc residues in the antennae

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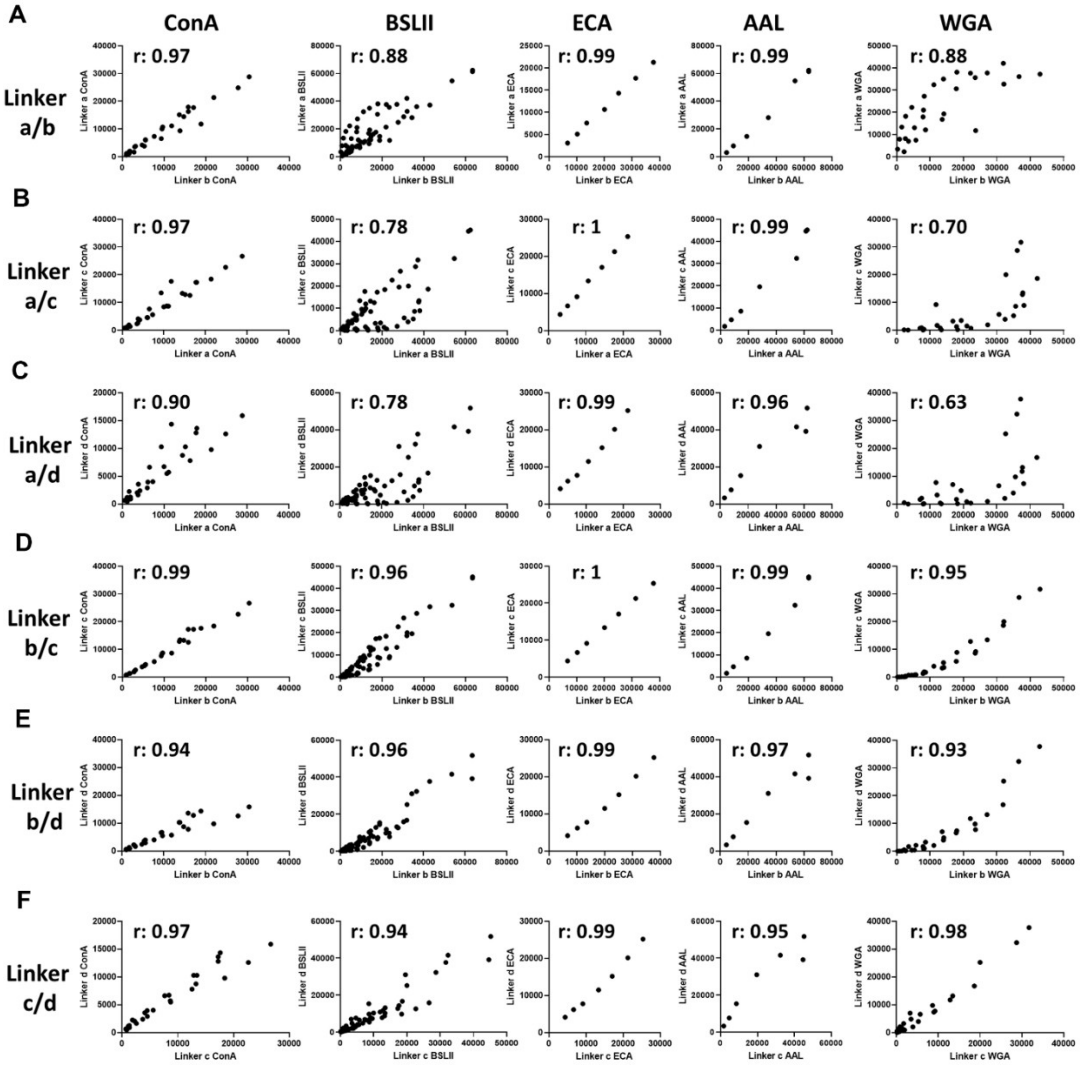


Figure 2. Comparison between the different linker pairs used in the glycan microarray. Scatter plot representation of the mean RFU values for six replicates spots for each linker are represented in x and y axes. Pearson correlation values *r* are plotted in each plot.

of the N-glycans with a slight preference for the α-1,3-arm.[44] For interaction with BSL-II lectin we found a lower correlation (*r* value of 0.78 for pairs **a**/**c** and **a**/**d**) between the aminopentyl linker **a** and the two peptidic linkers **c** and **d**. A similar trend was observed for wheat germ agglutinin presenting the lowest correlation values (*r* value of 0.70 for **a**/**c** pair and 0.63 for **a**/**d** pair) for the linkers used and thus highlighting a profound influence of the linker on the interaction with both lectins.

To compare the binding affinity between glycan-lectin pairs, we plotted the average RFU values for every glycan against the lectin concentrations and determined the surface dissociation constants ($K_{D\ surf}$) as previously described.[40] The results obtained are summarized in Table 2 (See Supporting Information for Langmuir isotherms representation). Additionally, a systematic comparison of the $K_{D\ surf}$ variability was determined by calculating the interquartile range (IQR) which is a measurement of the data dispersion. In Figure 3, a heat map represents the IQR values of the $K_{D\ surf}$ calculated for each lectin N-glycan pair immobilized through the four linkers in the microarray.

If we only analyse the RFU data at single protein concentration considering the different linkers employed, important differences in RFU intensities could be detected for some lectin-N-glycan pairs. Nevertheless, most of these differences are not appreciable when the dissociation constants ($K_{D\ surf}$) are calculated, resulting in similar IQR values for the $K_{D\ surf}$ for all four linkers employed in the immobilization. One example of this behaviour is observed in the interaction with *Concanavalin A* (ConA). We observed binding of ConA towards all N-glycans immobilized in the microarrays showing higher RFU values towards **M3** compounds (Figure 1A), possibly due an improved accessibility of the α-mannose residues which are the binding epitope for this lectin.[45] However, the RFU values for **M3d** glycan at different concentrations are reduced compared to **M3a**, **M3b** and **M3c**. When the $K_{D\ surf}$ values are calculated considering the whole range of lectin concentrations, all **M3** compounds showed similar dissociation constants on the surface in the nanomolar range (ranging from 16.9 to 24.3 nM, Table 2 and graphical representation in Supporting Information)

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Table 2. K_D surface [nM].

Lectin	ConA	BSLII	ECA	WGA	AAL
G0a	55.1 ± 5.6	1191 ± 118	nd	228 ± 84	nd
G0b	50.2 ± 5.9	1206 ± 124	nd	526 ± 154	nd
G0c	90.4 ± 6.7	1505 ± 39	nd	425 ± 29	nd
G0d	79.1 ± 3.0	3118 ± 269	nd	576 ± 83	nd
G2a	64.1 ± 6.7	nd	276 ± 34	368 ± 63	nd
G2b	82.7 ± 3.2	nd	239 ± 41	517 ± 121	nd
G2c	113.9 ± 18.6	nd	246 ± 37	1303 ± 422	nd
G2d	97.9 ± 5.4	nd	340 ± 74	683 ± 194	nd
M3a	16.9 ± 1.1	nd	nd	422 ± 61	nd
M3b	18.7 ± 1.9	nd	nd	2414 ± 607	nd
M3c	24.3 ± 2.2	nd	nd	5885 ± 918	nd
M3d	20.8 ± 4.3	nd	nd	5016 ± 1619	nd
G0Fa	49.4 ± 3.7	909 ± 36	nd	306 ± 41	2.9 ± 0.9
G0Fb	56.0 ± 7.2	1282 ± 68	nd	465 ± 39	2.4 ± 0.5
G0Fc	81.9 ± 7.4	2038 ± 176	nd	1020 ± 207	3.7 ± 1.0
G0Fd	126.7 ± 4.8	5248 ± 422	nd	955 ± 153	1.6 ± 0.7

nd: no determined. No binding detected.

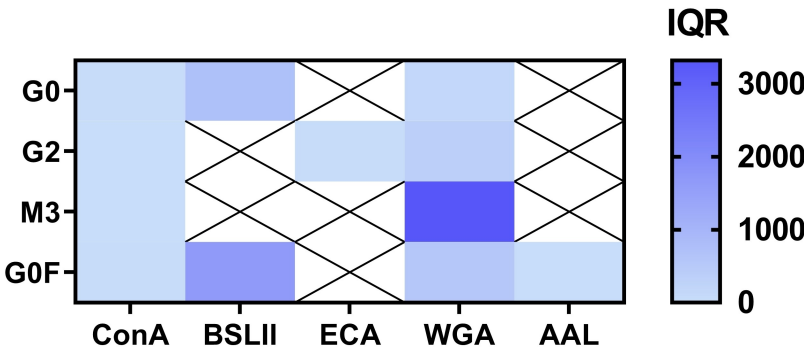


Figure 3. Heat map showing interquartile range (IQR) of the $K_{D\ surf}$ values for lectin binding to each N-glycan immobilized through the four linkers in the microarray. White wells with black cross means not determined, because there is no interaction with this lectin-N-glycan pairs.

and low variation in the IQR value (Figure 3). A similar behaviour was observed for *Erythrina cristagalli* agglutinin (ECA) which recognizes efficiently the Galβ1-4GlcNAc epitope presented in **G2** N glycan type structures. For all concentrations tested, **G2b** showed increased RFU values towards ECA but after calculation of affinity constants we could observed that all of them presented similar $K_{D\ surf}$ values (ranging from 239 and 340 nM, Table 2 and graphical representation in Supporting Information) in the higher nanomolar range and low variation in the IQR (Figure 3). These observations highlight the problem of ranking affinity glycan binding data obtained at a single lectin concentration which can lead to misleading results.

On the other hand, *Aleuria aurantia* lectin (AAL) recognized core fucosylated N-glycan structures **G0F** efficiently, with similar dissociation constants in the low nanomolar range independently of the linker used and with excellent Pearson correlation for the different linker pairs. The binding of AAL to α-1,6-core fucosylated N-glycans has been studied and shown to involve only contact points between the fucose residue and the lectin.[46] Despite the proximity of the fucose moiety to the anomeric position, we could not observe any influence of the linker on binding with this lectin.

Larger differences in IQR for the $K_{D\ surf}$, however, were observed for two of the lectins studied, wheat germ agglutinin (WGA) and *Griffonia (Bandeiraea) Simplicifolia* Lectin II (BSLII) (Figure 3), both lectins are known N-acetylglucosamine (GlcNAc) binding proteins. The $K_{D\ surf}$ values for each N-glycan with respect of the linker employed are represented in Figure 4. WGA is a 36 kDa lectin enriched from the seeds of *Triticum vulgaris* that binds to terminal N-acetylneuraminic acid (NeuAc), N-acetylglucosamine (GlcNAc) residues and GlcNAc oligosaccharides such as chitobiose (GlcNAcβ1-4GlcNAc), which are recognized with higher affinity compared to monomeric GlcNAc.[47] Due to the high affinity for the common chitobiose

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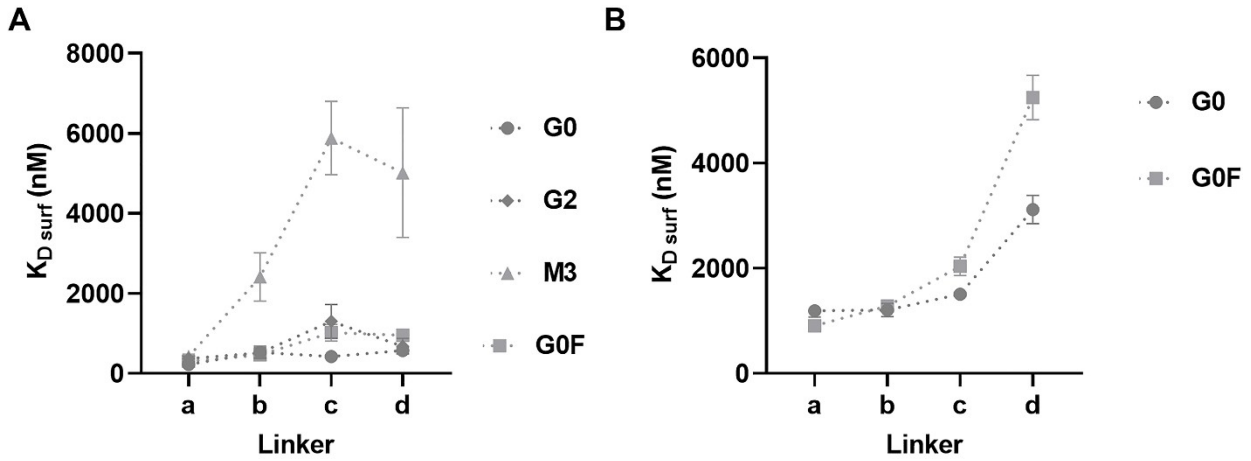


Figure 4. A. Binding of WGA towards N-glycan structures. $K_{D\ surf}$ for each N-glycan lectin pair is represented in function of the linker employed in the immobilization B. Binding of BSLII towards N-glycan structures. $K_{D\ surf}$ for each N-glycan lectin pair is represented in function of the linker employed in the immobilization.

core presented in all N-glycan structures, this lectin is widely used in glycomic studies for the enrichment/analysis of glycoproteins. For **M3** N-glycans, which could only be recognized via their chitobiose core, important differences could be seen depending on the linker employed. Considering the $K_{D\ surf}$ calculated on the microarray, **M3a** bound strongest, followed by **M3b**. **M3c** and **M3d** N-glycans containing "natural" peptidic linkers, interacted similarly but in a weaker interaction in the micromolar range (Figure 4A, See Supporting Information for curve representation). Previous solution NMR studies in complex N-glycans modified with Asn and peptide derived from egg yolk in the anomeric position, showed an exclusive interaction of WGA with the GlcNAc residue at the reducing end and the asparagine moiety, while no interaction was detected with the additional amino acids presented in the glycopeptide.[48]

The calculated surface binding constants $K_{D\ surf}$ for the interaction of *Griffonia (Bandeiraea) Simplicifolia* Lectin II (BSLII) with the printed glycans were in the low micromolar range and showed considerable variability. In particular, BSL II bound N-glycans immobilized via an anomeric asparagine ($K_{D\ surf}$ = 3.12 μM **G0d** and $K_{D\ surf}$ = 5.25 μM **G0Fd**) with a 2.5–4.3 lower binding affinity compared to the same N-glycans attached to the surface with linkers **a**, **b** and **c** ($K_{D\ surf}$ = 1.19 **G0a**, 1.21 **G0b**, 1.50 μM **G0c** and $K_{D\ surf}$ = 0.91 **G0Fa**, 1.28 **G0Fb**, 2.04 μM **G0Fc**). This decrease in binding might be caused by an unfavorable glycan surface presentation via this linker which is pronounced for the interaction with this lectin impeding the efficient recognition of the antennae.

Conclusions

We have presented a novel linker for the conjugation and isolation of free reducing glycans from natural sources that maintains a closed conformation of the reducing end GlcNAc residue, shows excellent reactivity in the conjugation of complex N-glycans and is equipped with a cleavable chromophoric group for purification and separation purposes. To further validate the linker for glycan array applications we produced a panel of synthetic N-glycan structures equipped with 4 different linkers, and performed microarray based binding studies with 5 different lectins. The new linker performed very well alongside more established methods of glycan immobilisation showing consistently superior binding in nearly all glycan-lectin interactions of this study. We conclude as have others before us, that glycan-GBP interactions on glycan microarrays can be considerably influenced by the nature of the linker used for the immobilization of glycans on the slide surface. An understanding of how the linker affects glycan presentation and ultimately lectin binding will help to develop more robust and meaningful glycan microarray platforms. Although efficient binding of the glycans was observed for all linkers employed for surface immobilisation, the binding between the constructs varied strongly when comparing single lectin concentrations. In addition, the binding affinity calculated as a surface dissociation constant $K_{D\ surf}$, however, showed a stronger linker dependence especially for weaker binders, while for high affinity binders with nanomolar binding constants the effect of the linker was far less visible. An exception at least in this limited study was WGA, which showed a pronounced linker dependant affinity with up to 14-fold variation in affinity among different glycoconjugates. A precise explanation for these observations cannot be given as the analysis of binding modes on immobilised substrates is not straightforward and a correlation with solution based binding models is likely to fail.[49] In the absence of molecular techniques for analysing the conformation of surface bound molecules, an explanation of the observed linker dependence remains largely speculative. Altogether our study shows that binding strengths observed at a single lectin concentration, which is a widespread practice for glycan microarray studies, can differ strongly for different linkers and even the calculated surface dissociation constants can very strongly vary for example, for lectins that show

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different binding modes. Consequently, binding data from glycan arrays employing a mix of different linkers for ligand immobilisation should be treated with caution, and similar effects are likely to occur for the presentation of synthetic glycans on beads and neoglycoproteins. Synthetic linkers like the urea type presented in this report are advantageous for the design of glycan arrays as they can be attached to any natural or synthetic glycan, maintain a close ring conformation at the reducing end sugar and avoid the mix of linkers and spacers often found in arrays composed of different glycoconjugate collections.

Experimental Section

Materials: Chemicals were purchased from Sigma-Aldrich or Acros Organics and were used without further purification. All organic solvents were dried over activated 4 Å or 3 Å molecular sieves. Thin layer chromatography was carried out using Merck aluminium sheets silica gel 60 F254 and visualized by UV irradiation (254 nm) or by staining with vanillin solution. Uridine 5′-diphosphogalactose disodium salt (UDP-Gal) and guanosine 5′-diphospho-β-L-fucose sodium salt (GDP-Fuc) were purchased from Carbosynth. Pronase® from *Streptomyces griseus* and β-galactosidase from *Aspergillus niger* were purchased from Sigma-Aldrich. Recombinant human core α-1,6-fucosyltransferase (FUT8) was purchased from ChemBind. Recombinant bovine milk β-1,4-galactosyltransferase was expressed in *E. coli* and folded from inclusion bodies as previously described.[50] Recombinant β-N-acetylglucosaminidase S from *Streptococcus pneumoniae* was purchased from New England Biolabs.

Instrumentation: Hydrogenation reactions were performed on a continuous-flow hydrogenation reactor H-Cube from ThalesNano Nanotechnology Inc. Purifications of compounds were performed on SampliQ high performance graphitized carbon cartridges from Agilent Technologies, C18 Sep-Pak Cartridges from Waters (Milford) and columns packed with BioGel P-4 (Biorad). All aqueous solutions were prepared from nanopure water produced with a Diamond UV water purification system (Branstead International). Pooled glycan containing fractions were lyophilized on an ALPHA-2-4 LSC freeze-dryer from Christ. All organic solvents were concentrated using rotary evaporation. NMR spectra were acquired on a Bruker 500 MHz spectrometer. Chemical shifts are reported in ppm (δ) and referenced to the residual signal of the solvent used (CD_3OD 4.87 ppm; $CDCl_3$ 7.26 ppm; D_2O 4.79 ppm). Splitting patterns are designated as s, singlet; d, doublet; t, triplet; m, multiplet. Coupling constants (*J*) are reported in Hz. MALDI-TOF mass analyses were performed on an Ultraflextreme III time-of-flight mass spectrometer equipped with a pulsed N_2 laser (337 nm) and controlled by FlexControl 3.3 software (Bruker Daltonics). The enzymatic reactions were heated in Eppendorf Thermomixer comfort and centrifuged in a bench microcentrifuge Eppendorf MiniSpin.

General procedure for N-glycan functionalization with linker 6

Deglycosylation of G1/G2-peptide using PNGaseF: G1/G2-peptide **7** (710 mg, 0.311 mmol) was dissolved in 4.5 mL of reaction buffer (20 mM Tris, 50 mM NaCl, 5 mM EDTA, pH = 7.5) and PNGaseF (800 U/μL, 200 μL) was added. The reaction mixture was incubated at 37 °C and the reaction progress was monitored via MALDI-TOF. Upon completion, the reaction mixture was heated to 75 °C for 15 min and the resulting mixture was treated with Pronase® for peptide degradation. The mixture was centrifuged at 4500 rpm for 10 min and the supernatant subjected to solid phase extraction using activated charcoal/Celite® column (1 : 1, 22 g:22 g). The column was eluted with H_2O containing 0.1 % TFA (500 mL), 25 % MeOH/H_2O containing 0.1 % TFA (1 L), 50 % MeOH/H_2O containing 0.1 % TFA (500 mL). The collected fractions were analyzed via MALDI-TOF spectrometry and fractions containing pure product were pooled, dried and lyophilized. N-glycan G1/G2 **8** was obtained as a white fluffy solid (482 mg, 0.294 mmol, 94 %).

Kochetkov reaction: Ammonium hydrogen carbonate was added to a solution of G1/G2 N-glycan **8** (200 mg, 155 μmol) in water until saturation. The reaction was stirred at 30 °C for 5 days. The solution was concentrated under reduced pressure and lyophilized. The compound **9** obtained was used for the next reaction without further purification.

Linker attachment: N-glycan **9** (200 mg, 122 μmol) was dissolved in 50 mL of dry DMSO containing DIPEA (106 μL, 608 μmol, 4 equiv.) and linker **6** (200 mg, 468 μmol, 4 equiv.) was added to the solution 3 times. The reaction was stirred reaction was stirred at r. t. for 24 h. The reaction progress was monitored via MALDI-TOF. DMSO was removed under reduced pressure and the crude product was purified via preparative HPLC on a Waters autopurification system including Waters 2767 Sample Manager collector, Waters 2996 Photodiode Array Detector, Waters SFO System Fluidics Organizer, Waters 2545 Binary Gradient Module, Waters 2475 Multi λ Fluorescence detector and Waters SQ detector 2 equipped with a standard ESI source (capillary voltage of 2.10 kV and a cone voltage of 50 V. Cone and desolvation gas flow were set to 50 and 900 L/h, respectively; source temperature and desolvation temperatures were 120 °C and 500 °C, respectively). Phenomenex Gemini RP C18 21x250 mm column (5 μm particle size) was employed in the purification. The samples were eluted with a flow rate of 16 mL/min, using a volume of injection of 100 μL and a maximum concentration 50 mg/mL of glycan in H_2O: ACN 8 : 2. The gradient employed has a mobile phase (A) 0.1 % TFA in H_2O/ (B) Acetonitrile; gradient: 0–3 at min 80 % A, 3–33 min to 45 % A, 33–33.50 min to 20 % A, 33.50-36.50 min at 20 % A, 36.50-37 min to 80 % A and 37–40 min at 80 % A. Linker functionalized N-glycan **10** was isolated as a white solid (165 mg, 84.5 μmol, 69 %).

Linker deprotection: Protected N-glycan **10** (40 mg, 21.0 μmol) was dissolved in H_2O/MeOH (3 mL, 1 : 1 containing 1 % TFA) and subjected to catalytic hydrogenation in H-cube® using a 10 % Pd/C cartridge (flowrate: 1 mL/min, full H_2, 50 °C). The product was analyzed via MALDI-TOF and the solvents removed under reduced pressure. The crude reaction was dissolved in water (10 mL) and extracted with toluene (3x10 mL). The aqueous phase was evaporated under reduced pressure and freeze dried to yield the linked N-glycan **11** as a white solid (35 mg, 21.0 μmol, 99 %).

General procedures of enzymatic modifications of N-glycans

Galactosylation (***Enzymatic reaction 1***): The linked **G1/G2** compound (2 equiv.) and uridine 5′-diphosphogalactose (1 equiv.) were dissolved in 0.5 mL buffer solution (50 mM Tris-HCl, 20 mM $MnCl_2$, pH 7.5). Bovine milk β-1,4-galactosyltransferase (2.5 U/mL) was added to the reaction. The reaction mixture was incubated at 37 °C for 24 h. The reaction progress was monitored via MALDI-TOF. The crude was centrifuged at 12500 rpm for 5 min, and the supernatant was purified by BioGel P-4 column chromatography with water as the eluent. After lyophilization, pure **G2** N-glycan was obtained as colourless solid.

Degalactosylation (***Enzymatic reaction 2***): The linked **G1/G2** compound (1 equiv.) was dissolved in 0.5 mL of NaOAc/AcOH buffer (100 mM, pH 4.5). β-galactosidase (*Aspergillus niger*, 5 U/mL) was added, and the reaction mixture incubated at 37 °C for 24 h.

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The reaction progress was monitored via MALDI-TOF. When full conversion was observed, the reaction was heated at 90 °C during 15 min and centrifuged. The supernatant was purified by solid phase extraction in carbon graphitized cartridge (2 g) equilibrated with acetonitrile and water. Elution was performed with a gradient of acetonitrile (0% to 100%) in water. Product eluted in 10%–25% acetonitrile containing fractions. Glycan containing fractions were pooled, evaporated and freeze dried to yield pure **G0** N-glycan as colourless solid.

Core fucosylation (*Enzymatic reaction 3*): G0 N-glycan (1 equiv.) and guanosine 5′-diphospho-β-L-fucose (2.5 equiv.) were dissolved in 0.9 mL Tris-HCl buffer (100 mM, pH 7.4). Recombinant α-(1,6)-fucosyltransferase from *Homo Sapiens* (FUT8) (20 μL, 0.24 U/mL) was added to the reaction. The reaction was incubated at room temperature for 96 h. The reaction progress was monitored via MALDI-TOF. The supernatant was purified by solid phase extraction in carbon graphitized cartridge (2 g) equilibrated with acetonitrile and water. Elution was performed with a gradient of acetonitrile (from 0% to 100%) in water. Product eluted in 10%–25% acetonitrile containing fractions. Glycan containing fractions were pooled, evaporated and freeze dried to yield pure G0F.

GlcNAc hydrolysis (*Enzymatic reaction 4*): G0 N-glycan (1 equiv.) was dissolved in 0.5 mL NaOAc/AcOH buffer (50 mM, pH 5.5) containing $CaCl_2$ (20 mM). β-N-acetylglucosaminidase S (*Streptococcus pneumoniae*, 10 U/μmol of G0) was added. The reaction mixture was incubated at 37 °C and the reaction progress was monitored via MALDI-TOF. After 24 h, MeOH (250 μL) was added and the reaction mixture was centrifuged at 12500 rpm for 5 min. The organic solvent was removed under reduced pressure and the crude product was purified by BioGel P-2 column chromatography with water as the eluent. Glycan containing fractions were pooled and freeze dried to yield **M3** N-glycans.

G2b was synthesized following the protocol of enzymatic reaction 1. From 7.5 mg of **G1/G2b**, 5.89 mg (3.6 μmol, 83%) of **G2b** were obtained as a white fluffy solid. ^{1}H NMR (D_2O, 500 MHz): δ 5.12 (s, 1H, H-$1_{Man\text{-}2}$), 4.93 (s, 1H, H-$1_{Man\text{-}3}$), 4.89 (d, J = 9.5 Hz, 1H, βH-$1_{GlcNAc\text{-}1}$), 4.61 (d, J = 8.0 Hz, 1H, H-$1_{GlcNAc\text{-}2}$), 4.58 (d, J = 7.7 Hz, 2H, H-$1_{GlcNAc\text{-}3}$, H-$1_{GlcNAc\text{-}4}$), 4.47 (dd, J = 7.9, 3.0 Hz, 2H, H-$1_{Gal\text{-}1}$, H-$1_{Gal\text{-}2}$), 4.25 (s, 1H, H-$2_{Man\text{-}2}$), 4.19 (s, 1H, H-$2_{Man\text{-}3}$), 4.11 (s, 1H, H-$2_{Man\text{-}1}$), 4.01–3.47 (m, 50H), 3.12 (t_{app}, J = 6.9 Hz, 2H, CH_2, H-5), 2.95 (t, J = 7.4 Hz, 2H, CH_2, H1), 2.08 (s, 3H $CH_{3,GlcNAc}$), 2.05 (s, 3H, $CH_{3,GlcNAc}$), 2.04 (s, 3H $CH_{3,GlcNAc}$), 2.012 (s, 3H $CH_{3,GlcNAc}$), 1.67–1.61 (m, 2H, CH_2, H-4), 1.53–1.47 (m, 2H, CH_2, H-2), 1.39–1.34 (m, 2H CH_2, H-3). HRMS (MALDI TOF): *m/z* calculated for $C_{68}H_{117}N_7O_{46}Na$: 1790.6920 $[M+Na]^+$; found 1790.6854.

G0b was synthesized following the protocol of enzymatic reaction 2. From 6.6 mg of **G2b**, 4.9 mg (3.4 μmol, 91%) of **G0b** were obtained as a white fluffy solid. ^{1}H NMR (D_2O, 500 MHz): δ 5.12 (s, 1H, H-$1_{Man\text{-}2}$), 4.92 (s, 1H, H-$1_{Man\text{-}3}$), 4.89 (d, J = 9.6 Hz, 1H, βH-$1_{GlcNAc\text{-}1}$), 4.77 (s, 1H, H-$1_{\beta Man}$), 4.61 (d, J = 7.7 Hz, 1H, H-$1_{GlcNAc\text{-}2}$), 4.55 (d, J = 8.4 Hz, 2H, H-$1_{GlcNAc\text{-}3}$, H-$1_{GlcNAc\text{-}4}$), 4.25 (s, 1H, H-$2_{Man\text{-}2}$), 4.19 (s, 1H, H-$2_{Man\text{-}3}$), 4.11 (s, 1H, H-$2_{Man\text{-}1}$), 3.98–3.41 (m, 40H), 2.97 (t, J = 6.6 Hz, 2H, CH_2, H5), 2.97–2.95 (m, 2H, CH_2, H1), 2.08 (s, 3H, $CH_{3,\ GlcNAc}$), 2.05 (s, 6H, $CH_{3,\ GlcNAc}$), 2.01 (s, 3H, $CH_{3,\ GlcNAc}$), 1.69–1.63 (m, 2H, CH_2, H-4), 1.53–1.47 (m, 2H, CH_2, H-2), 1.44–1.34 (m, 2H, CH_2, H-3). HRMS (MALDI TOF): *m/z* calculated for $C_{56}H_{97}N_7O_{36}Na$: 1466.5863 $[M+Na]^+$; found 1466.5838.

G0Fb was synthesized following the protocol of enzymatic reaction 3. From 3.95 mg of **G0b**, 3.71 mg (2.3 μmol, 86%) of **G0Fb** were obtained as a white fluffy solid. ^{1}H NMR (D_2O, 500 MHz): δ 5.12 (s, 1H, H-$1_{Man\text{-}2}$), 4.92 (s, 1H, H-$1_{Man\text{-}3}$), 4.87 (m, 2H, H-1_{Fuc}, H-$1_{GlcNAc\text{-}1}$), 4.78 (s, 1H, H-$1_{Man\text{-}1}$), 4.69 (d, J = 7.9 Hz, 1H, H-$1_{GlcNAc\text{-}2}$) 4.56 (d, J = 8.4 Hz, 2H, H-$1_{GlcNAc\text{-}3}$, H-$1_{GlcNAc\text{-}4}$), 4.25 (s, 1H, H2-$_{Man2}$), 4.19 (s, 1H, H2-$_{Man3}$), 4.12-4.10 (s, 1H, H2-$_{Man1}$), 3.98–3.44 (m, 50H), 3.11 (t, J = 6.8 Hz, 2H, CH_2, H5), 2.96 (t, J = 7.6 Hz, 2H, CH_2, H1), 2.09–2.01 (m, 12H, $CH_{3,GlcNAc}$), 1.68–1.62 (m, 2H, CH_2, H-4), 1.52–1.47 (m, 2H CH_2, H-2), 1.39–1.32 (m, 1H, CH_2, H-3), 1.20 (d, J = 6.6 Hz, 3H, H-6_{Fuc}). HRMS (MALDI TOF): *m/z* calculated for $C_{62}H_{107}N_7O_{40}Na$: 1612.6442 $[M+Na]^+$; found 1612.6520.

M3b was synthesized following the protocol of enzymatic reaction 4. From 2.9 mg of **G0b**, 1.9 mg (2.5 μmol, 95%) of **M3b** were obtained as a white fluffy solid. ^{1}H NMR (D_2O, 500 MHz): δ 5.11 (s, 1H, H-$1_{Man\text{-}2}$), 4.92 (s, 1H, H-$1_{Man\text{-}3}$), 4.89 (d, J = 9.5 Hz, 1H, H-$1_{GlcNAc\text{-}1}$), 4.61 (d, J = 8.2 Hz, 1H, H-$1_{GlcNAc\text{-}2}$), 4.26 (s, 1H, H-$2_{Man\text{-}2}$), 4.07 (dd, J = 3.4, 1.7 Hz, 1H, H-$2_{Man\text{-}3}$), 3.98 (dd, J = 3.5, 1.7 Hz, 1H, H-$2_{Man\text{-}1}$), 3.91–3.59 (m, 27H), 3.12–3.10 (m, 2H, CH_2, H5), 2.98–2.95 (m, 2H, CH_2, H1) 2.08 (s, 3H, $CH_{3,\ GlcNAc}$), 2.01 (s, 3H, $CH_{3,\ GlcNAc}$), 1.69–1.63 (m, 2H, CH_2, H4), 1.55–1.47 (m, 2H, CH_2, H2), 1.40–1.34 (m, 2H, CH_2, H3). HRMS (MALDI TOF): *m/z* calculated for $C_{40}H_{71}N_5O_{26}Na$: 1060.4276 $[M+Na]^+$; found 1060.4284.

Glycan microarray fabrication. Solutions of glycans (~1.5 nL) at 50 μM concentration diluted in 300 mM sodium phosphate buffer containing 0.005% Tween®-20 were spatially arrayed on Nexterion® H (Schott AG) slides employing a piezoelectric non-contact printer (SciFLEXARRAYER S11, Scienion). After overnight incubation in a humidity chamber, the unreacted NHS groups were quenched by immersion in ethanolamine (50 mM) in sodium borate buffer (50 mM) pH 9.0. The slides were washed with water and stored at −20 °C until use.

The microarrays were compartmentalized with ProPlate® microarray system (24 wells) and 100 μL of each fluorescent labelled lectin in binding buffer (25 mM Tris, 150 mM NaCl, 2 mM $CaCl_2$, 2 mM $MgCl_2$ and 0.001% Tween®-20) was applied and incubated in the dark for 1 h. The slides were washed with binding buffer followed with water and dried in a slide spinner. The slides were scanned on a microarray scanner (Agilent G2565BA, Agilent Technologies) at 10 μm resolution. Quantification of fluorescence was performed by ProScanArray® Express software (Perkin Elmer). Mean RFU values with local background subtraction of six spots and standard deviation of the mean were represented in histograms employing GraphPad Prism® software. Mean RFU values and lectin concentrations were represented as saturation curves and $K_{D\ surf}$ were calculated adjusting the values to non-linear curve fitting: RFU = (RFUmax*[lectin])/($K_{D\ surf}$ + [lectin]) in GraphPad Prism® software.

Author Contributions

R.A. and S.S. designed the experiments and analyzed the data. R.A., S.S. and D.P. prepared the compounds and carried out the microarray experiments. R.A. and S.S. interpreted the data. R.A., S.S., R. L. and N.R wrote the manuscript. All authors have read and approved the manuscript.

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Conflict of Interests

N.-C.R. is CEO and shareholder of Asparia Glycomics S.L., a company commercializing glycoscience products and services.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: carbohydrate-protein interactions · glycan microarrays · lectins · linker · N-glycans

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